EVAN ALEKSEYEV

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Edge AI & Embedded Systems engineer with 1 year of experience writing CV pipelines and production firmware

Education

San Jose State University — San Jose, CA
Bachelor of Science in Computer Engineering — *Expected May 2026, 3.6 GPA*

Technical Skills

Languages: Python, C, C++, x86/ARM Assembly, Verilog HDL
ML & Inference: PyTorch, OpenCV, TensorRT, ONNX Compilation, OpenVINO, Jetson Orin, Model quantization
Tools: Linux, Git, Docker, LangGraph, scikit-learn, PostgreSQL, SQLite, LanceDB
Embedded & Hardware: TI MSPM0, TI IWR68xx radar, CAN Bus, LIN, UART, SPI, ESP32, DeGirum Orca1

Experience

Software Engineer — March 2026 – Present
DeGirum Corp. — *Santa Clara, CA*

- Built production vehicle ReID pipeline (YOLO + OSNet + PaddleOCR) on live RTSP streams with proximity-zone alerting; deployed on DeGirum Orca1 edge accelerator
- Designed DAG execution runtime enabling LLMs to compile and orchestrate structured tool-call graphs dynamically
- Integrated face ReID and object detection into a LangGraph agentic VMS to run on multiple camera feeds

Mechatronics Engineer — Sep 2025 – March 2026
Beagle Technology, Inc — *Dublin, CA*

- Contributed to R&D and field deployment of prototype agricultural robotics products, shipping computer vision improvements and PID tuning directly to automated machinery at customer sites
- Sole firmware owner (C++): rewrote I/O PCB codebase to support 4 PCB revisions, 2 machine types, and 12-channel relay + 5 channel proportional valve control; designed CAN bus communication layer to NVIDIA Jetson
- Built remote MJPEG streaming service and fleet monitoring dashboard with GPS tracking, & real-time device health; deployed query optimizations reducing load times by 90%

Academic Projects

Texas Instruments Sponsored Capstone — August 2025 – May 2026
Texas Instruments — *Santa Clara, CA*

- Served as scrum master & engineering lead for senior capstone project, with mentorship from TI engineers
- Architected multi-node radar fusion pipeline normalizing IWR6843AOP track data (position, velocity, SNR) across 3 distributed sensor nodes into standardized JSON packets for cloud transmission

Hobby Projects

E90 Remote Start – BMW CAN Bus Controller | *C++, ESP32, CAN Bus, WiFi/HTTP* — 2025

- ESP32-based CAN bus controller interfacing with BMW E90 K-CAN network for real-time telemetry and control
- Reverse engineered 20+ CAN IDs covering engine RPM, throttle, climate, central locking, and remote start
- Designed custom hardware to interface with vehicle start/stop system for remote start functionality

Add Spice - 3D CNN Video Analysis | *Python, PyTorch, OpenCV, 3D CNNs* — 2025

- Designed a neural network (DinoV2 backbone + temporal transformer embedder + dense head with RankNet loss) to objectively rate intensity 0–10 from GoPro clips for real-time evaluation on RTX 3050
- Built dataset annotation environment and end-to-end training pipeline with augmentation and checkpointing

ShakespeareLM – Transformer Language Model | *Python, PyTorch, Transformers* — 2025

- Built a decoder-based language model in PyTorch, trained on Shakespeare, Kafka, and Dostoevsky corpuses
- Achieved basic generalization; model internalized proper grammatical and punctuation conventions